

Requirements Table — Vehicle Parking System

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall display available parking slots and allow the user to select a nearby slot.	High	Pass: Available slots are displayed and the selected slot number is provided. Fail: Available slots or slot number is incorrect/not displayed.	Helps users quickly find parking without searching manually.
FR-002	Functional	The system shall allow customers to reserve an available parking slot for a specified time.	High	Pass: The selected slot is reserved for the specified time and shown as reserved. Fail: The slot is not reserved correctly.	Ensures parking availability for customers in advance.
FR-003	Functional	The system shall record the vehicle number and entry time, assign a parking slot, and mark the slot as occupied.	High	Pass: Vehicle details and entry time are recorded, and the assigned slot is marked occupied. Fail: Any vehicle or parking information is incorrect or missing.	Enables accurate vehicle tracking and parking management.
FR-004	Functional	The system shall calculate the parking fee based on entry and exit time and update the vehicle and slot status after payment.	High	Pass: Correct fee is calculated, payment is recorded, vehicle is marked exited, and slot becomes available. Fail: Fee or status is incorrect.	Ensures accurate payment and timely availability of parking slots.
FR-005	Functional	The system shall display the number of occupied and available slots and allow the manager to mark slots as unavailable for maintenance.	High	Pass: Current parking status is displayed and maintenance slots can be marked unavailable. Fail: Status is incorrect or slots cannot be updated.	Helps the manager monitor and maintain the parking facility.
NFR-001	Non-Functional	The system shall display the current parking availability within 2 seconds of the customer's request.	High	Pass: Availability is displayed within 2 seconds. Fail: The response takes more than 2 seconds.	Provides a quick and efficient user experience.
NFR-002	Non-Functional	The system shall restrict access to parking management and	High	Pass: Only authorized staff can access management and payment details. Fail:	Protects sensitive parking and

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		payment information to authorized staff only.		Unauthorized users can access restricted information.	payment information.